

[illegible]

1.1 □□□□

Model	SAITS	SAITS+PatchTST	Model
A SAITS	144	137	SAITS+PatchTST (SAITS+PatchTST)
B SAITS	64	59	SAITS+PatchTST (50%/70% SAITS+PatchTST)
C SAITS	48	48	SAITS+PatchTST (SAITS+PatchTST)
D SAITS	24	21	SAITS+PatchTST (SAITS+PatchTST)
E SAITS	64	56	SAITS+PatchTST (SAITS+PatchTST)
F SAITS	12	12	SAITS+PatchTST (SAITS+PatchTST)
Total	356	333	

1.2 实验目的

333/356 23 Traffic862+ CRIB-full CRIB patching + MultivariateNormal 862 batch_size=4 ~72GB epoch 504 8.4 3.5 23 40+

CRIB-full Traffic MSE > 0.9 CRIB

1.3 □□□□□□□□

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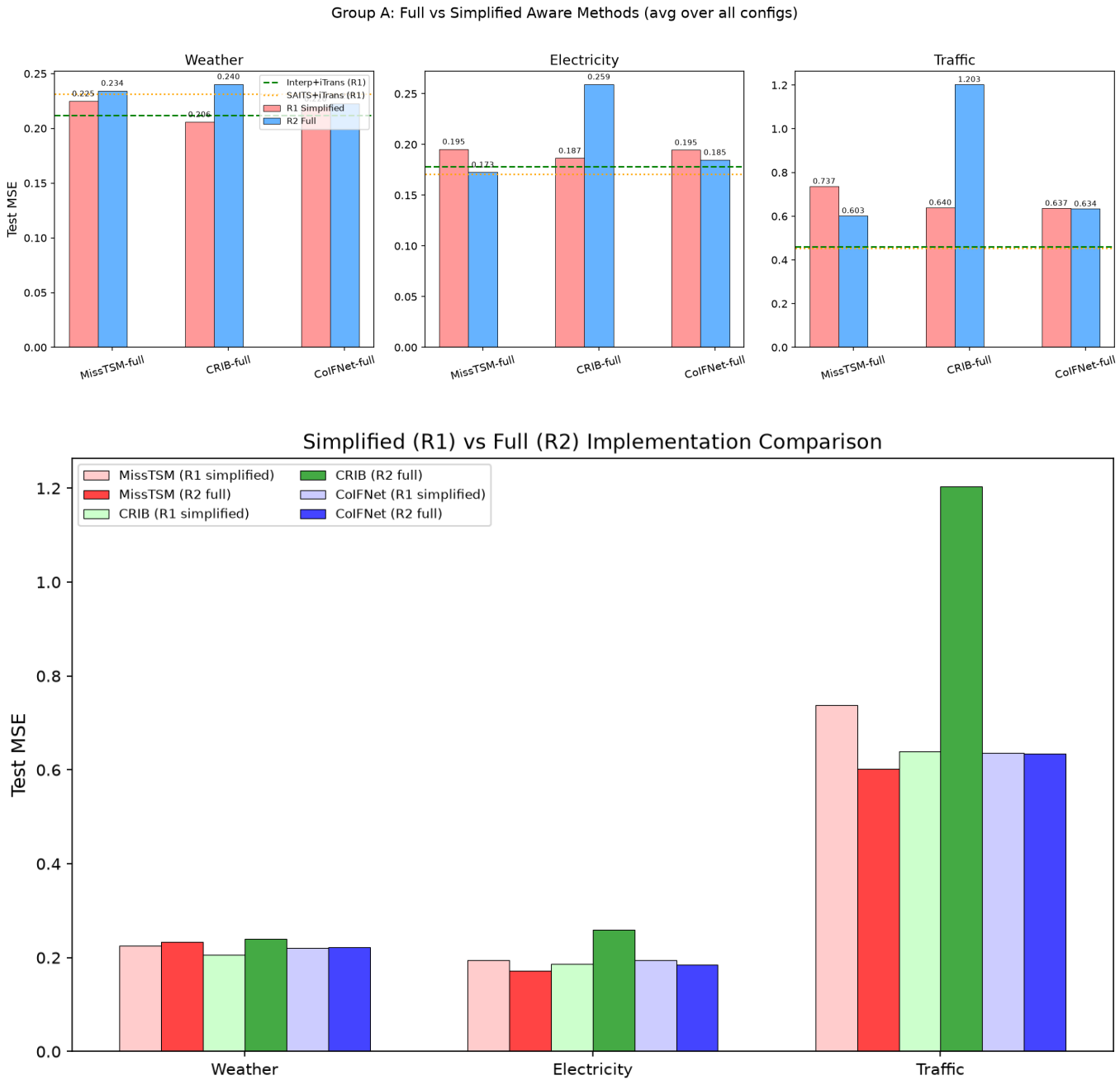
Model	Model	Model	Model
Model	Model CRIB/CoIFNet/MissTSM	Model	Model
Model	10 epoch, patience=3	20 epoch, patience=5	Model
Model	0.1, 0.3	0.5, 0.7	Model
Model	3	2	Model

0000 A00000——000000000000 vs 000000

2.1 00000

0000 MissTSM/CRIB/CoIFNet 000000000000Interp+iTrans 0 SAITS+iTrans00000000000000000000

000000



Weather

0000	000	00	Interp+iTrans (R1)	SAITS+iTrans (R1)	MissTSM- full	CRIB-full	CoIFNet- full
0000	10%	96	0.1761 /	0.1754 /	0.1786 /	0.1778 /	0.1696 /
0			0.2179	0.2279	0.2448	0.2155	0.2176

00000	0000	00 00	Interp+iTrans (R1)	SAITS+iTrans (R1)	MissTSM- full	CRIB-full	ColFNet- full
00000 0	10%	336	0.2746 / 0.2964	0.2918 / 0.3309	0.3000 / 0.3477	0.2952 / 0.3162	0.2804 / 0.3025
00000 0	30%	96	0.1742 / 0.2187	0.1779 / 0.2425	0.1728 / 0.2389	0.1793 / 0.2192	0.1696 / 0.2177
00000 0	30%	336	0.2709 / 0.2967	0.2902 / 0.3396	0.2884 / 0.3342	0.3107 / 0.3329	0.2762 / 0.3007
00000 00	10%	96	0.1755 / 0.2170	0.1676 / 0.2135	0.1783 / 0.2409	0.1691 / 0.2095	0.1667 / 0.2139
00000 00	10%	336	0.2798 / 0.2988	0.2785 / 0.3062	0.2910 / 0.3351	0.2947 / 0.3158	0.2741 / 0.2990
00000 00	30%	96	0.1876 / 0.2285	0.1788 / 0.2367	0.1778 / 0.2392	0.1805 / 0.2202	0.1656 / 0.2111
00000 00	30%	336	0.2887 / 0.3050	0.2898 / 0.3281	0.2877 / 0.3313	0.3146 / 0.3337	0.2788 / 0.3025

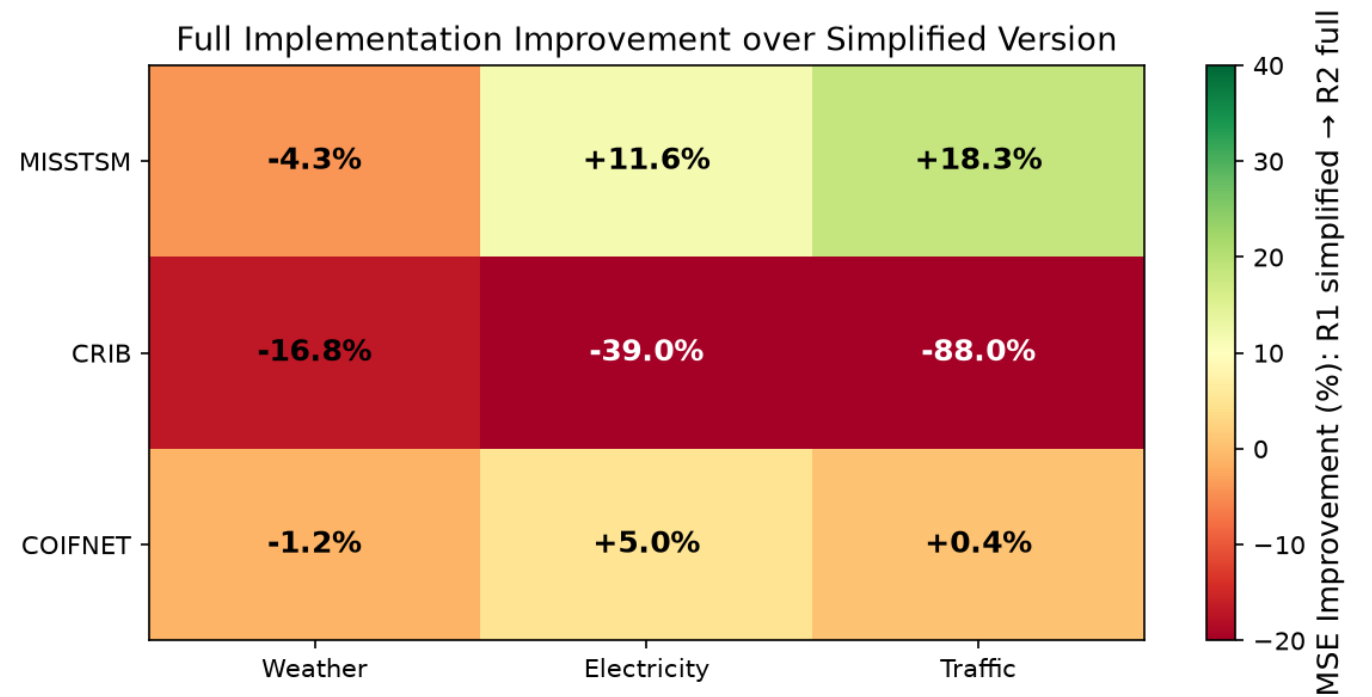
Electricity

00000	0000	00 00	Interp+iTrans (R1)	SAITS+iTrans (R1)	MissTSM- full	CRIB-full	ColFNet- full
00000 0	10%	96	0.1579 / 0.2521	0.1519 / 0.2500	0.1490 / 0.2515	0.1542 / 0.2496	0.1668 / 0.2709
00000 0	10%	336	0.1873 / 0.2804	0.1807 / 0.2791	0.1877 / 0.2891	0.1839 / 0.2795	0.1941 / 0.2962
00000 0	30%	96	0.1636 / 0.2603	0.1608 / 0.2649	0.1520 / 0.2567	0.1595 / 0.2562	0.1697 / 0.2750
00000 0	30%	336	0.1902 / 0.2862	0.1845 / 0.2894	0.1850 / 0.2892	0.5272 / 0.5265	0.1956 / 0.3001
00000 00	10%	96	0.1648 / 0.2579	0.1567 / 0.2517	0.1547 / 0.2543	0.1545 / 0.2485	0.1648 / 0.2691
00000 00	10%	336	0.1975 / 0.2891	0.1813 / 0.2763	0.1900 / 0.2885	0.1841 / 0.2782	0.1985 / 0.3009
00000 00	30%	96	0.1860 / 0.2760	0.1630 / 0.2625	0.1625 / 0.2653	0.1711 / 0.2626	0.1813 / 0.2828
00000 00	30%	336	0.2200 / 0.3075	0.1878 / 0.2871	0.2014 / 0.3032	0.5397 / 0.5326	0.2086 / 0.3084

Traffic

Model	Series	Size	Interp+iTrans (R1)	SAITS+iTrans (R1)	MissTSM- full	CRIB-full	ColFNet- full
MissTSM	Weather	10%	0.4237 / 0.2954	0.4253 / 0.2940	0.6495 / 0.3792	0.9095 / 0.5417	0.5998 / 0.3285
MissTSM	Electricity	10%	0.4565 / 0.3068	0.4559 / 0.3055	0.6069 / 0.3442	0.9342 / 0.5512	0.6492 / 0.3426
MissTSM	Traffic	30%	0.4373 / 0.3023	0.4374 / 0.2970	0.5775 / 0.3305	1.4141 / 0.8019	0.6068 / 0.3389
MissTSM	Weather	30%	0.4687 / 0.3135	0.4691 / 0.3105	0.6066 / 0.3372	1.4332 / 0.8082	0.6374 / 0.3477
CRIB	Weather	10%	0.4455 / 0.3071	0.4296 / 0.2940	0.6162 / 0.3609	-	0.6101 / 0.3329
CRIB	Electricity	10%	0.4779 / 0.3191	0.4643 / 0.3076	0.6043 / 0.3412	1.4315 / 0.8075	0.6503 / 0.3444
CRIB	Traffic	30%	0.4912 / 0.3294	0.4515 / 0.3001	0.5743 / 0.3287	-	0.6470 / 0.3519
ColFNet	Weather	30%	0.5233 / 0.3420	0.4929 / 0.3197	0.5870 / 0.3262	1.4499 / 0.8124	0.6741 / 0.3619

2.2 R1 vs R2



Model	Series	R1 MSE	R2 MSE	Improvement
MissTSM	Weather	0.2247	0.2343	-4.3%

Model	Category	R1 MSE	R2 MSE	Delta
MissTSM	Electricity	0.1954	0.1728	+11.6%
MissTSM	Traffic	0.7375	0.6028	+18.3%
CRIB	Weather	0.2058	0.2402	-16.8%
CRIB	Electricity	0.1866	0.2593	-39.0%
CRIB	Traffic	0.6400	1.2034	-88.0%
ColFNet	Weather	0.2200	0.2226	-1.2%
ColFNet	Electricity	0.1946	0.1849	+5.0%
ColFNet	Traffic	0.6369	0.6343	+0.4%

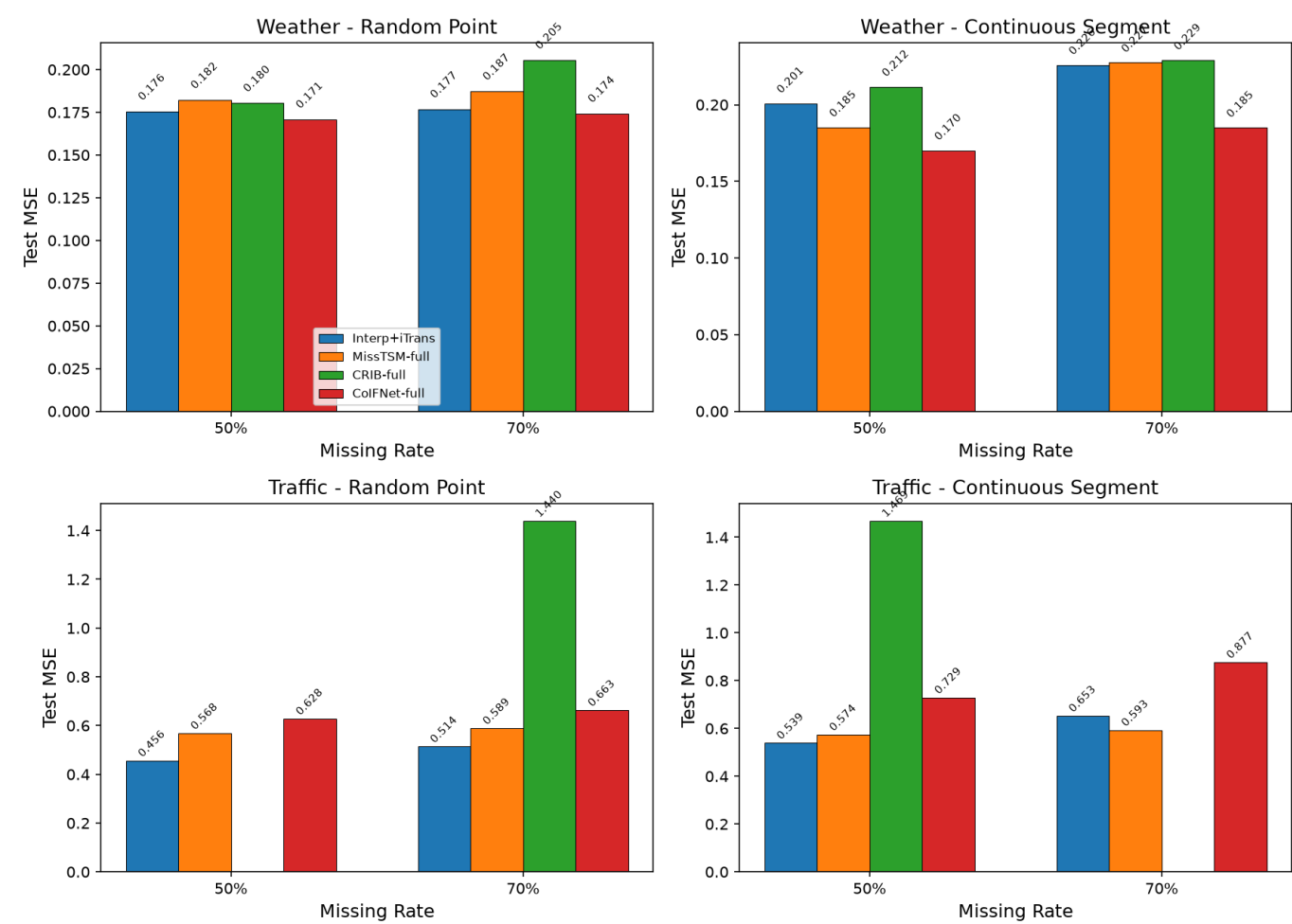
2.3 A

- 1. **MissTSM-full Electricity** MissTSM-full Electricity 30%/h96: 0.1520 vs Interp+iTrans 0.1636 7.1% skip connection Traffic 18.3%
- 2. **CRIB-full** CRIB Electricity Traffic -39%-88%Electricity h336 MSE 0.53R1 0.19Traffic MSE > 0.9 MultivariateNormal 20 epoch
- 3. **ColFNet-full** Weather -1.2%Electricity +5%Traffic +0.4% ColFNet
- 4. Interp+iTransMSE 0.2834 SAITS+iTrans0.2851 MissTSM-full0.3366 ColFNet-full0.3473 19-22%

B50%/70%

3.1

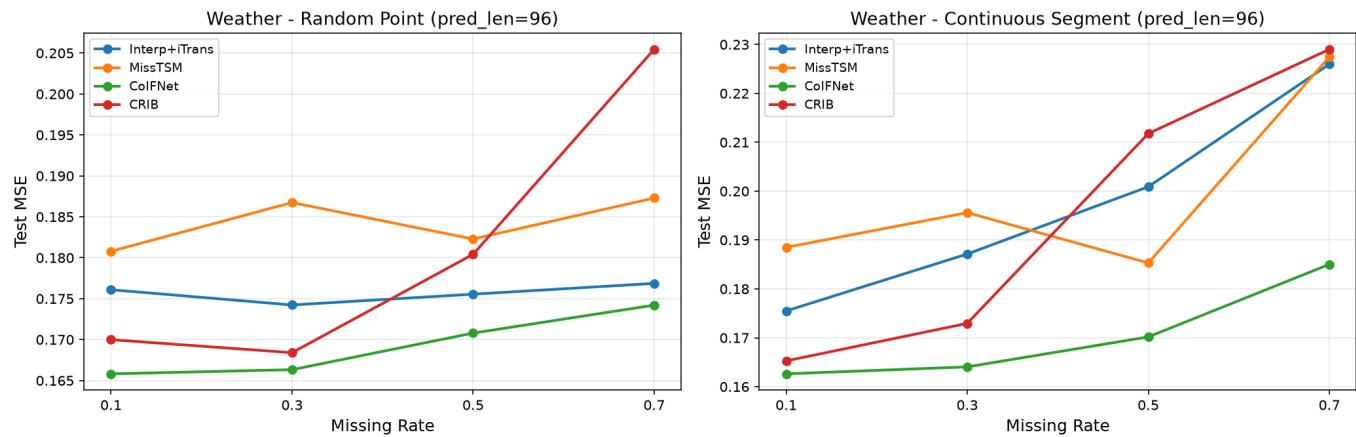
Group B: High Missing Rate Exploration (50% & 70%)



Dataset	Query Type	Missing Rate	Interp+iTrans	MissTSM-full	CRIB-full	CoIFNet-full
Weather	Random Point	50%	0.1755 / 0.2211	0.1823 / 0.2490	0.1804 / 0.2206	0.1708 / 0.2192
Weather	Random Point	70%	0.1769 / 0.2258	0.1873 / 0.2521	0.2055 / 0.2512	0.1742 / 0.2215
Weather	Continuous Segment	50%	0.2009 / 0.2401	0.1853 / 0.2520	0.2118 / 0.2537	0.1702 / 0.2145
Weather	Continuous Segment	70%	0.2260 / 0.2620	0.2275 / 0.2811	0.2290 / 0.2697	0.1850 / 0.2241
Traffic	Random Point	50%	0.4562 / 0.3094	0.5681 / 0.3217	-	0.6278 / 0.3458
Traffic	Random Point	70%	0.5139 / 0.3372	0.5888 / 0.3341	1.4398 / 0.8092	0.6631 / 0.3692
Traffic	Continuous Segment	50%	0.5389 / 0.3431	0.5742 / 0.3256	1.4687 / 0.8161	0.7290 / 0.3918
Traffic	Continuous Segment	70%	0.6529 / 0.3908	0.5930 / 0.3308	-	0.8768 / 0.4752

3.2 0000000010% → 70%

Missing Rate Trend: 10% → 70% (Weather)



3.3 B

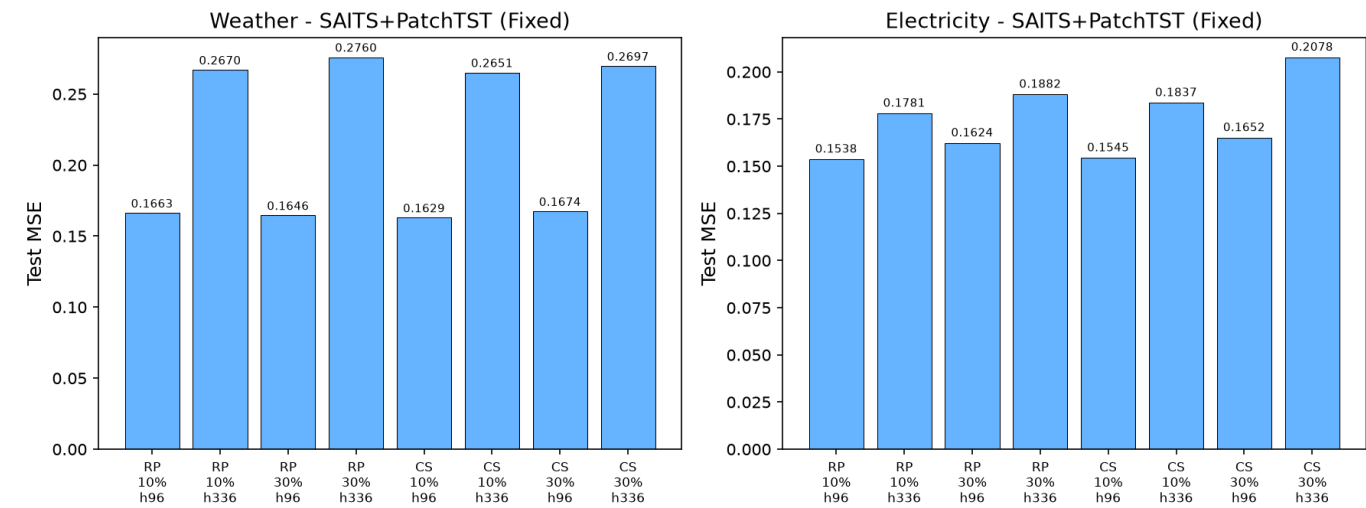
- 1. **ColFNet-full** Weather Weather 4 ColFNet-full MSE0.1702~0.1850 Interp+iTrans0.1755~0.2260 2.7%~18.1%
- 2. **MissTSM-full** Traffic 70% Traffic, 70% MissTSM-full0.5930 Interp+iTrans0.6529 9.2%"
- 3. **Interp+iTrans** Weather 0.187630% 0.226070% 20.5% ColFNet-full 0.165630% 0.185070% 11.7%
- 4. **CRIB-full** Traffic MSE > 1.4 OOM
- 5. Weather 30~50% ColFNet-full

C SAITS+PatchTST

4.1

SAITS+PatchTST Weather/Electricity 102 20 epoch, patience=5 48

Group C: SAITS+PatchTST Overflow Fix Results



Dataset	Model	Horizon	H=96 MSE/MAE	H=336 MSE/MAE
Weather	SAITS+PatchTST	10%	0.1657 / 0.2098	0.2495 / 0.2800
Weather	SAITS+PatchTST	30%	0.1646 / 0.2136	0.2502 / 0.2809
Weather	SAITS+PatchTST	10%	0.1643 / 0.2084	0.2480 / 0.2764
Weather	SAITS+PatchTST	30%	0.1709 / 0.2169	0.2500 / 0.2830
Electricity	SAITS+PatchTST	10%	0.1481 / 0.2403	0.1734 / 0.2696
Electricity	SAITS+PatchTST	30%	0.1531 / 0.2487	0.1779 / 0.2737
Electricity	SAITS+PatchTST	10%	0.1517 / 0.2429	0.1719 / 0.2680
Electricity	SAITS+PatchTST	30%	0.1640 / 0.2597	0.1930 / 0.2879

4.2 C

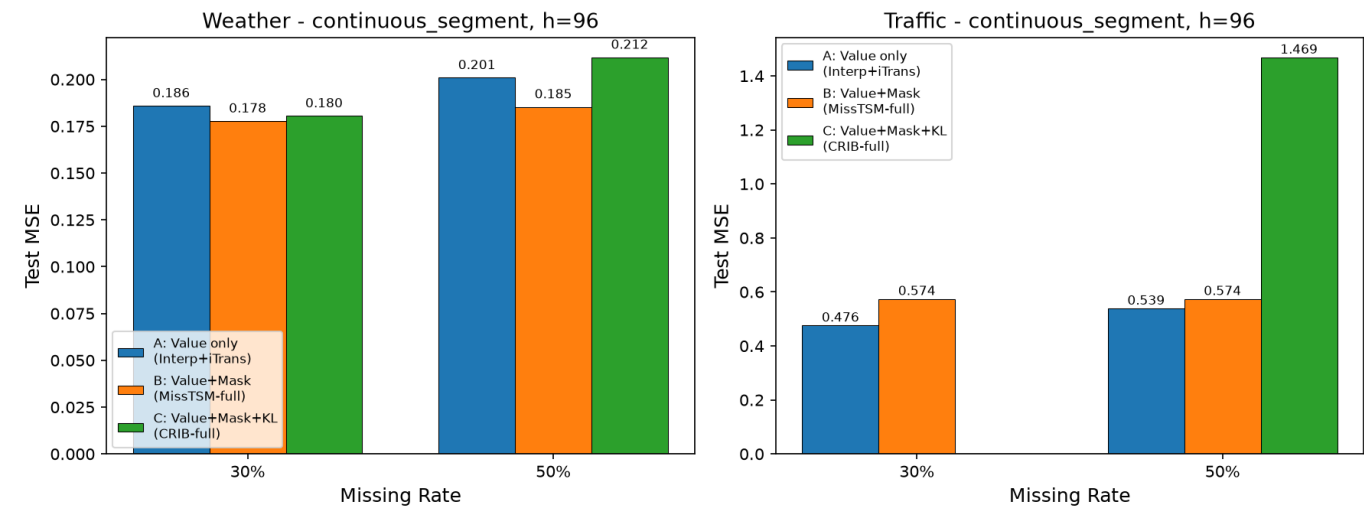
SAITS+PatchTST Electricity h96 MSE 0.1481~0.1640 SAITS+iTransformer MissTSM-full Weather h96 MSE 0.1643~0.1709

D

5.1

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Group D: Mask Ablation with Full-version Methods



		30% Test MSE	50% Test MSE
Weather	A: Interp+iTrans	0.1876	0.2009
Weather	B: MissTSM-full	0.1778	0.1853
Weather	C: CRIB-full+KL	0.1805	0.2118
Traffic	A: Interp+iTrans	0.4912	0.5389
Traffic	B: MissTSM-full	0.5743	0.5742
Traffic	C: CRIB-full+KL	-	-

5.2 D

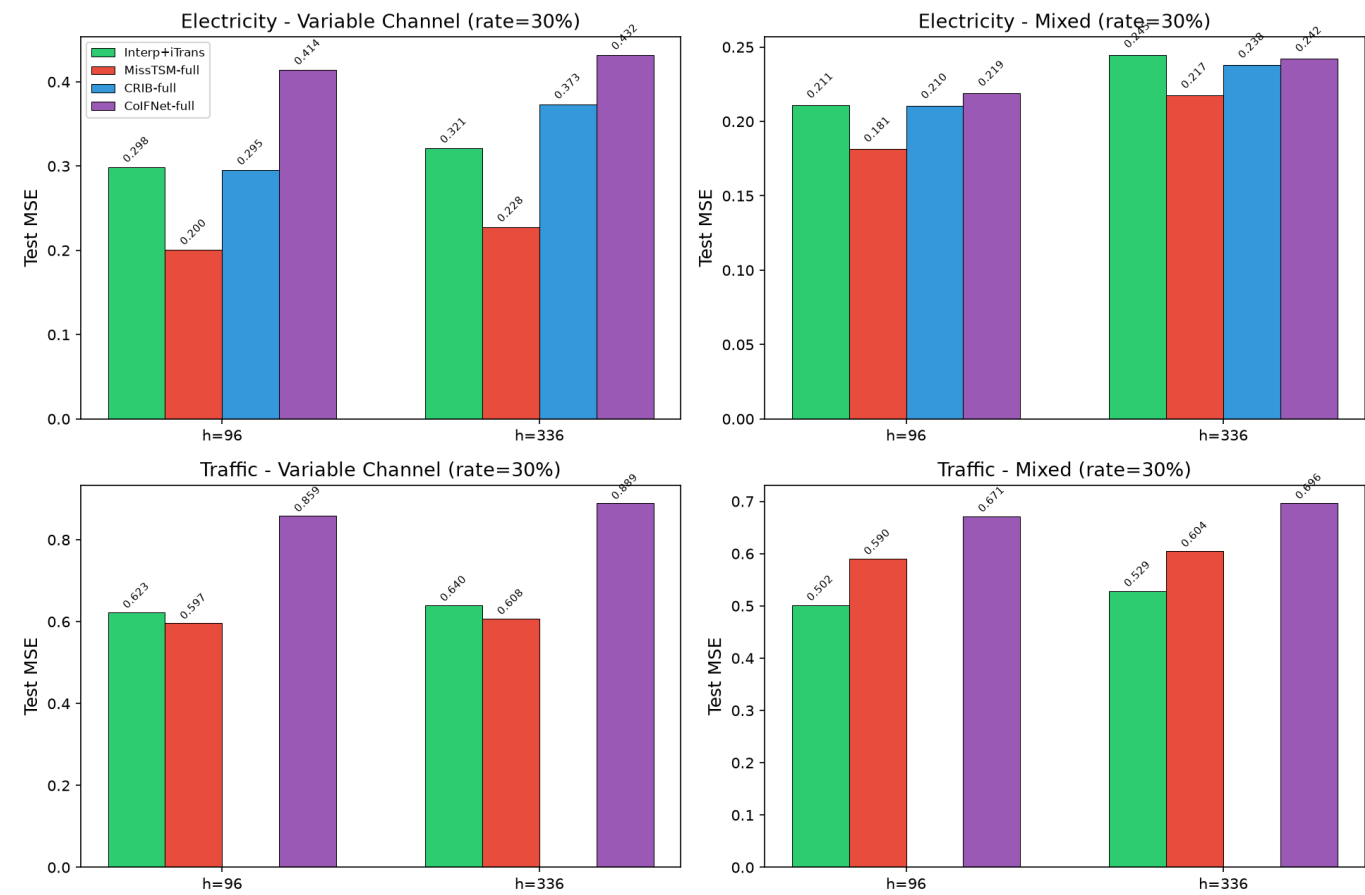
- **Weather** MissTSM-full0.1778/0.1853 Interp+iTrans0.1876/0.2009 50% 7.8% MissTSM skip connection
- **Traffic** MissTSM-full0.5743 Interp+iTrans0.4912

E

6.1

Electricity Traffic

Group E: Extended Missing Scenarios (variable_channel & mixed)



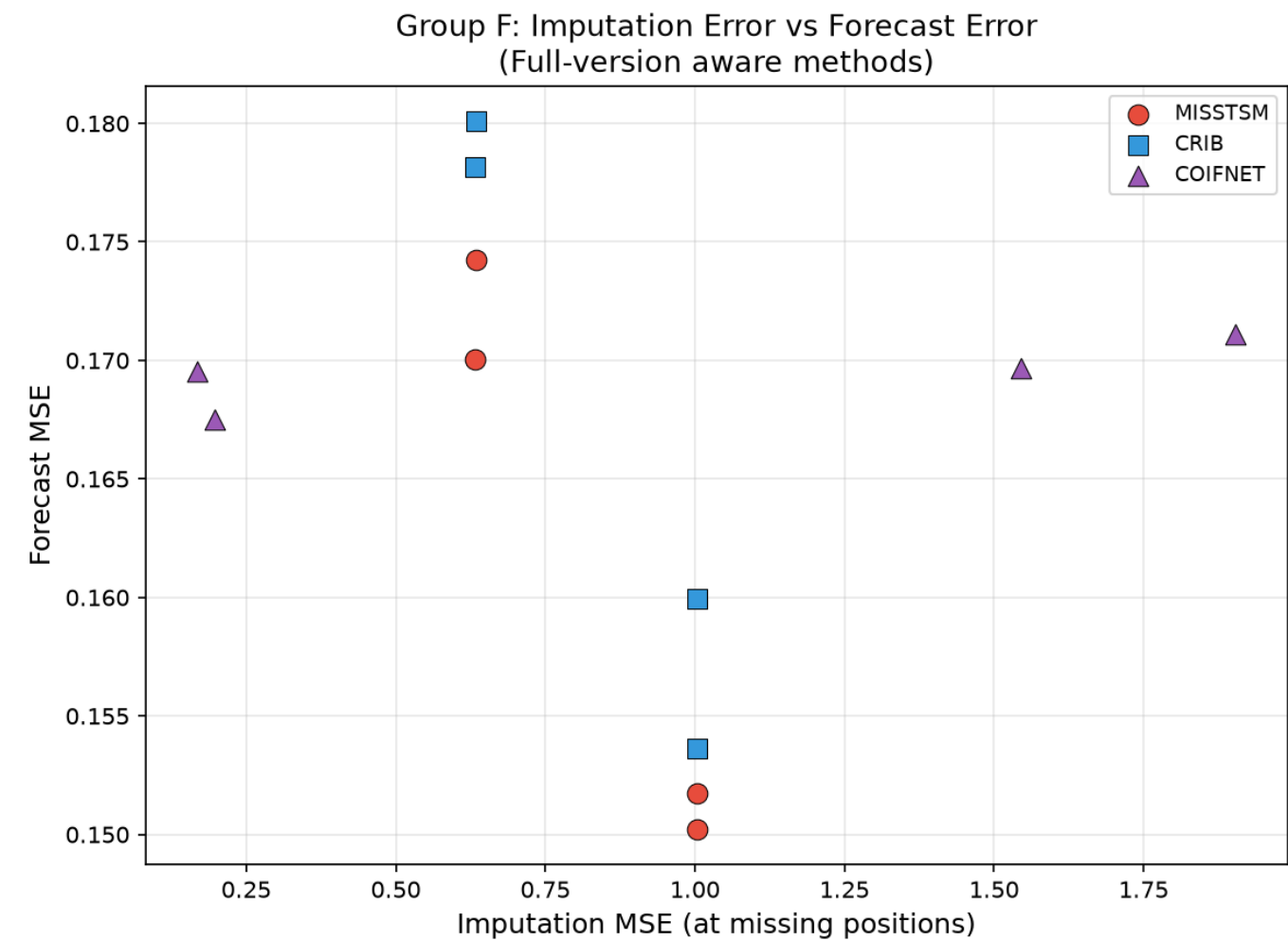
Dataset	Scenario	h	Interp+iTrans	MissTSM-full	CRIB-full	ColFNet-full
Electricity	Variable Channel	96	0.1730 / 0.2673	0.1842 / 0.2777	0.1850 / 0.2776	0.1949 / 0.2940
Electricity	Variable Channel	336	0.2060 / 0.2952	0.2154 / 0.3082	0.2182 / 0.3079	0.2290 / 0.3222
Electricity	Mixed	96	0.1694 / 0.2643	0.1764 / 0.2735	0.1780 / 0.2719	0.1884 / 0.2887
Electricity	Mixed	336	0.2009 / 0.2922	0.2124 / 0.3044	0.2106 / 0.3046	0.2217 / 0.3179
Traffic	Variable Channel	96	0.5030 / 0.3233	0.6078 / 0.3428	-	0.7271 / 0.3800
Traffic	Variable Channel	336	0.5260 / 0.3338	0.6119 / 0.3423	-	0.8199 / 0.4221
Traffic	Mixed	96	0.4860 / 0.3180	0.5984 / 0.3437	-	0.7183 / 0.3863
Traffic	Mixed	336	0.5191 / 0.3324	0.5999 / 0.3398	-	0.8072 / 0.4239

6.2 E

Interp+iTrans

F

7.1



Model	Category	Missing %	Imputation MSE	Forecast MSE
MissTSM-full	Weather	10%	0.6340	0.1742
MissTSM-full	Weather	30%	0.6328	0.1700
MissTSM-full	Electricity	10%	1.0038	0.1502
MissTSM-full	Electricity	30%	1.0038	0.1517
CRIB-full	Weather	10%	0.6340	0.1801
CRIB-full	Weather	30%	0.6328	0.1781
CRIB-full	Electricity	10%	1.0038	0.1536
CRIB-full	Electricity	30%	1.0038	0.1599
CoIFNet-full	Weather	10%	0.1970	0.1675
CoIFNet-full	Weather	30%	0.1687	0.1695
CoIFNet-full	Electricity	10%	1.5454	0.1697
CoIFNet-full	Electricity	30%	1.9034	0.1711

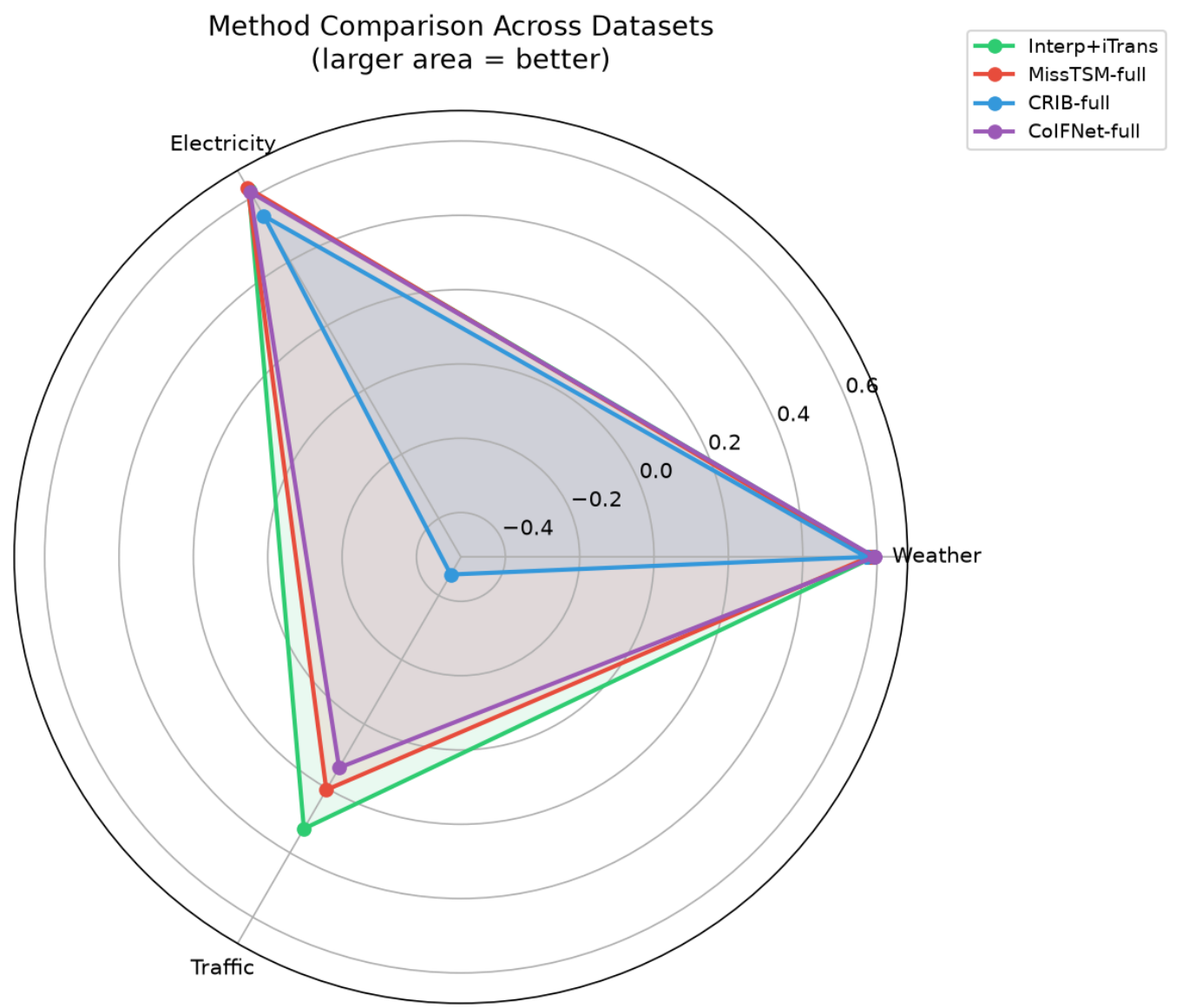
7.2 F

- CoIFNet-full 的 MSE 在 Electricity 上 1.5~1.9 倍于 MissTSM-full 的 MSE，即 $MSE_{CoIFNet} \neq MSE_{MissTSM}$
- MissTSM-full 在 Electricity 上的 MSE 为 0.1502，而 CRIB-full 的 MSE 为 1.0

实验结果

8.1 实验结果

实验结果在 MSE 上 A 组 3 组 × 2 组 × 2 组 × 2 组



Rank	Method	MSE	Relative Error
1	Interp+iTrans (R1)	0.2834	—
2	SAITS+iTrans (R1)	0.2851	+0.6%
3	MissTSM-full (R2)	0.3366	+18.8%
4	CoIFNet-full (R2)	0.3473	+22.5%
5	CRIB-full (R2)	0.4591	+62.0%

8.2

F1: +iTransformer		
F2: >		
F3: MissTSM-full Electricity CoIFNet-full Weather		
F4: ≠	F	
F5: MissTSM Weather Traffic		

8.3

MissTSM/CoIFNet:	MissTSM Electricity 11.6% Traffic 18.3%	
CoIFNet: Weather	CoIFNet-full Weather 50%/70%	
MissTSM: Traffic 70%	Traffic 70% MissTSM Interp+iTrans	
	CRIB-full	

8.4

1. CRIB-full patching + MultivariateNormal +
- Traffic MSE 0.64 1.20 88%

Electricity h336 MSE0.53 MultivariateNormal

Traffic epoch 504 batch_size 4
2. MissTSM skip connection MissTSM $x = mask * x_{inp} + (1 - mask) * out$ Electricity 11.6%
3. Weather 30~50% CoIFNet-full Traffic CoIFNet-full Traffic

9.1

1. MissTSM+11~18% CoIFNet+0~5% CRIB -17~88%

